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Development and Simulation of a Whole-Body Control Framework for a Humanoid Robot

Master's Thesis in Robotics Engineering

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2026

Abstract

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Keywords: Whole-Body Control, Humanoid Robot, Quadratic Programming, Centroidal Dynamics, Simulation, Loco-manipulation, Task-Priority Control.

Acknowledgements

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Chapter 1

Introduction

1.1 Motivation

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Critical Insight

1.2 Problem Statement

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Critical Insight

1.3 Research Question and Objectives

> *"How can a Whole-Body Control (WBC) framework be designed and validated in simulation to enable a humanoid robot to perform stable, multi-contact locomotion and manipulation tasks under dynamic constraints?"*

The following sub-questions structure the investigation:

SQ1 What are the key limitations of existing WBC formulations (task-priority, QP-based, MPC) when applied to full-body humanoid coordination?

SQ2 How can contact dynamics be modeled and integrated in a simulation environment to faithfully replicate hardware behavior?

SQ3 What metrics and scenarios constitute a rigorous simulation-based validation of a WBC framework for humanoids?

1.4 Hypotheses

Hypothesis H1

A hierarchical Quadratic Program (QP)-based Whole-Body Control (WBC) formulation can achieve stable whole-body coordination across locomotion and manipulation tasks simultaneously.

Hypothesis H2

Simulation-based validation (Isaac Sim / MuJoCo) is sufficient to demonstrate proof-of-concept for the proposed WBC framework before hardware deployment.

Hypothesis H3

Explicit contact modeling and task-priority hierarchies outperform naive torque control in multi-contact transitions.

1.5 Scope and Limitations

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1.6 Contributions

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1.7 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews the state of the art in whole-body control for humanoid robots, covering model-based methods, centroidal MPC, and learning-based baselines. Chapter 3 formalizes the proposed WBC framework and justifies the design choices. **??** describes the simulation environment, robot model, and software stack. Chapter 5 presents quantitative results across evaluation scenarios S1 through S5. Chapter 6 interprets the results and discusses limitations. Chapter 7 summarizes contributions and outlines future research directions.

Chapter 2

State of the Art: Whole-Body Control for Humanoid Robots

Chapter Outline

This chapter synthesises the model-based Whole-Body Control (WBC) literature for humanoid robots under three framing constraints inherited from the thesis master memory. First, the review prioritises peer-reviewed contributions from IEEE, Springer, and Elsevier venues, supplemented by recent (post-2022) preprints when they fill identifiable gaps. Second, the taxonomy is preserved across the chapter: methods are grouped as task-priority / null-space, QP-based, MPC-based, and (as baseline reference) learning-based. Third, the chapter terminates with an explicit positioning of the proposed thesis contribution against the canvassed body of work, in line with the mandated logical flow.

The chapter is therefore not a chronological catalogue but an *argument*: the dominant architectural pattern (planning layer plus execution layer) is well-established but underspecified at the coupling interface, and the absence of a standardised evaluation protocol for simultaneous loco-manipulation constitutes a credible target for a master's-level contribution.

Critical Insight

The structural choice to organise the chapter around an argument rather than a method-by-method survey carries a methodological cost: it requires the reader to accept the taxonomy as load-bearing before the evidence is presented. This risk is mitigated by reserving Section 2.7 for the synthesis and by explicitly flagging, throughout, where method boundaries blur (e.g. between QP-WBC and MPC at the centroidal level). A reviewer may legitimately object that the taxonomy elides hybrid model-learning approaches; this chapter treats those as out of scope per the thesis scope statement, while Section 2.5 acknowledges them as a reference baseline.

2.1 Foundations of Whole-Body Control

Whole-Body Control (WBC) emerged at the intersection of operational-space control [**?**] and floating-base rigid-body dynamics, motivated by the recognition that a humanoid robot's degrees of freedom cannot be commanded independently without violating either dynamic feasibility or contact-stability constraints [**?**]. The defining problem WBC addresses is the *simultaneous coordination of a high-DOF system* (typically 28–32 DOF for humanoids: torso, arms, and legs) under a hierarchy of behavioural objectives, while respecting the physical constraints imposed by joint limits, actuator torque saturation, and unilateral contact wrenches [**?** **?**].

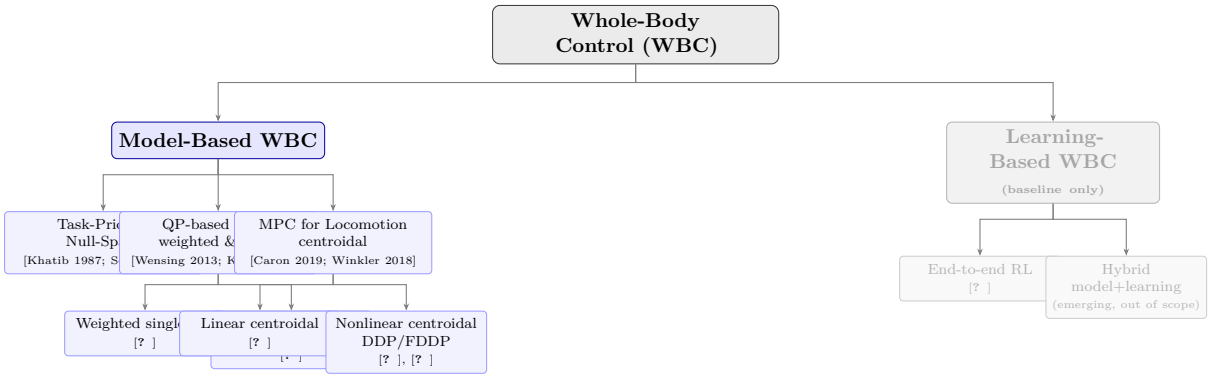

Figure 2.1: Taxonomy of Whole-Body Control (WBC) approaches for humanoid robots. Model-based methods (blue) are the primary focus of this chapter; learning-based approaches (grey) are retained as baseline references only (see Section 2.5). Representative works are indicated in brackets.

The canonical formulation considers a humanoid as a *floating-base* system: six unactuated degrees of freedom describe the base pose relative to an inertial frame, augmented by n actuated joint variables. The governing equation of motion takes the standard form

$$\boldsymbol{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{S}^\top \boldsymbol{\tau} + \boldsymbol{J}_c^\top \boldsymbol{f}_c \tag{2.1}$$

where $\boldsymbol{M}(\boldsymbol{q})$ denotes the mass matrix, $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q})$ collects the Coriolis, centrifugal, and gravity terms, $\boldsymbol{S}$ is the actuation selection matrix, $\boldsymbol{\tau}$ collects the actuated joint torques, and $(\boldsymbol{J}_c, \boldsymbol{f}_c)$ are the contact Jacobian and the stacked contact wrench respectively [? ? ?]. WBC frameworks differ primarily in *how* they reconcile multiple, potentially conflicting task objectives over (2.1) while ensuring that $\boldsymbol{f}_c$ lies inside the friction cone.

Three foundational design choices have shaped the field. The first is the choice of *control level*: kinematic (velocity), inverse-dynamic (acceleration), or torque. Torque-level WBC, although computationally heavier, is generally regarded as the most expressive level for handling contact-rich behaviours [?]. The second is the choice of *task representation*: end-effector position, momentum, centroidal quantities, or arbitrary task-space coordinates [?]. The third is *how priorities between tasks are enforced* — through null-space projection (Section 2.2), through weighted optimisation (Section 2.3), or through hierarchical optimisation with strict priorities in Hierarchical Quadratic Program (QP) form [? ? ? ?].

Critical Insight

The foundational WBC formulation already exposes the central trade-off that recurs throughout the literature: *constraint satisfaction* versus *computational tractability*. Torque-level WBC with hard friction-cone constraints is mathematically rigorous but compute-intensive; kinematic-level WBC with weighted priorities is fast but elides the contact physics. The thesis's choice to operate at QP-torque level inherits the former trade-off and must justify the additional cost in Chapter 2's successor: a justification that rests on the rigorous constraint handling that later sections will document.

2.2 Task-Priority and Null-Space Methods

The earliest Whole-Body Control (WBC) formulation that generalised cleanly to floating-base humanoids is the task-priority null-space approach introduced by Sentis and Khatib [?]. Their hierarchical control of behavioural primitives expresses each task as a desired acceleration in a chosen task space; lower-priority tasks are projected into the null space of the higher-priority Jacobians $\boldsymbol{J}$, ensuring that they are executed *only insofar as they do not perturb* higher-priority objectives. This formulation has the appealing property of *strict* hierarchy: the priority ordering is enforced exactly, not approximately.

Subsequent work refined this framework along three axes. The first axis is *smoothness at task transitions*, where pseudo-inverse switching can produce velocity discontinuities — addressed by continuous null-space projectors [?]. The second is the *handling of singularities*, where Jacobian rank deficiencies require regularisation or task reactivation strategies. The third is the *integration of inequality constraints*: classical null-space methods natively handle equality constraints (e.g. desired task accelerations) but offer no clean mechanism for inequality constraints such as friction cones, joint limits, or torque bounds [?].

Empirically, null-space WBC has been demonstrated on humanoid systems including the original DLR-Justin and a range of simulated platforms, achieving task hierarchies of moderate depth (typically three to five priority levels) [?]. Its computational cost is favourable — projector chains can be evaluated in milliseconds for high-DOF systems — but the absence of an inequality-constraint mechanism is broadly recognised as a fundamental limitation when contact stability or torque limits become binding [? ?].

Critical Insight

Null-space WBC is best understood as an *expressive but partial* solution: it answers the question "how do I order tasks?" rigorously, but cannot answer "how do I respect physical bounds while doing so?" without auxiliary mechanisms. The QP-based reformulation discussed in Section 2.3 emerged precisely to absorb both questions into a single optimisation problem. A reviewer might argue, fairly, that null-space methods are not obsolete — they remain attractive when constraints are slack and compute is at a premium — but the debate is context-dependent and unresolved. The thesis takes a principled position toward QP-WBC while refraining from dismissing null-space methods uncritically.

2.3 QP-based Whole-Body Control Formulations

The Quadratic Program (QP) formulation reframes the Whole-Body Control (WBC) problem as a constrained optimisation solved at every control cycle [?]. Decision variables typically include joint accelerations $\ddot{\boldsymbol{q}}$, contact wrenches $\boldsymbol{f}_c$, and (optionally) torques $\boldsymbol{\tau}$; the cost function is a weighted sum of task tracking errors, commonly expressed through norms $\|\boldsymbol{e}\|$ over task-space error signals; and the constraints encode the rigid-body dynamics (2.1), joint and torque limits, and contact-friction conditions [? ?]. The first contribution to formalise this approach with simultaneous satisfaction of ZMP and friction constraints at all priority levels was Wensing and Orin's 2013 paper on conic optimisation for dynamic humanoid behaviours [?], which demonstrated dynamic kicks and jumps on a 26-DOF simulated humanoid with a roughly 2× computational speed-up over prior task-space methods.

Three subsequent lines of work consolidated QP-WBC as the model-based standard. First, *momentum-based formulations* [?] reframed the cost around centroidal momentum tracking and were deployed on the Atlas humanoid at the DARPA Robotics Challenge, with foot-placement errors reported up to 5 cm during rough-terrain walking [?]. Second, *Hierarchical QP (HQP)* formulations introduced strict priorities into the QP cost, generalising the null-space approach of Section 2.2 to inequality-constrained settings [?]. Third, whole-body admittance control with Quadratic Program (QP) wrench distribution extended QP-WBC to industrial multi-contact tasks such as stair climbing on the HRP-4 platform [?].

Recent post-2022 work has sharpened the focus on *computational robustness and platform scaling*. Liu et al. [?] propose a model-based HQP formulation tailored to humanoids with heavy limbs, where standard formulations become numerically ill-conditioned. Wang et al. [?] couple an Model Predictive Control (MPC) layer with a whole-body planner-controller for underactuated humanoids, and Smaldone et al. [?] show that explicit attention to the compute budget at the planner/WBC interface yields robust dynamic walking. Across these works, QP solve times remain in the *sub-millisecond regime* when implemented with tuned solvers such as Eiquadprog or PROXQP [?].

A key methodological observation is that *friction-cone compliance* in QP-WBC is rarely reported as a percentage in the literature, because the QP enforces it as a hard inequality constraint: violations correspond to solver infeasibility events, not to gradual non-compliance. This observation is revisited in Section 2.7.

Critical Insight

QP-WBC is the field's mature technical answer to the multi-task, multi-constraint problem, but its maturity hides a measurement problem: most papers report constraint *satisfaction* qualitatively (the QP was feasible) rather than quantitatively (what fraction of cycles, under what perturbation budget). The post-2022 literature has begun to expose this gap [? ?] but has not yet converged on a reporting protocol — a gap Section 2.7 will argue the thesis can credibly target. A second insight is that the boundary between QP-WBC and MPC is increasingly porous: at the centroidal level, the same dynamics is being optimised in both, just over different horizons. This blurring is the subject of Section 2.4.

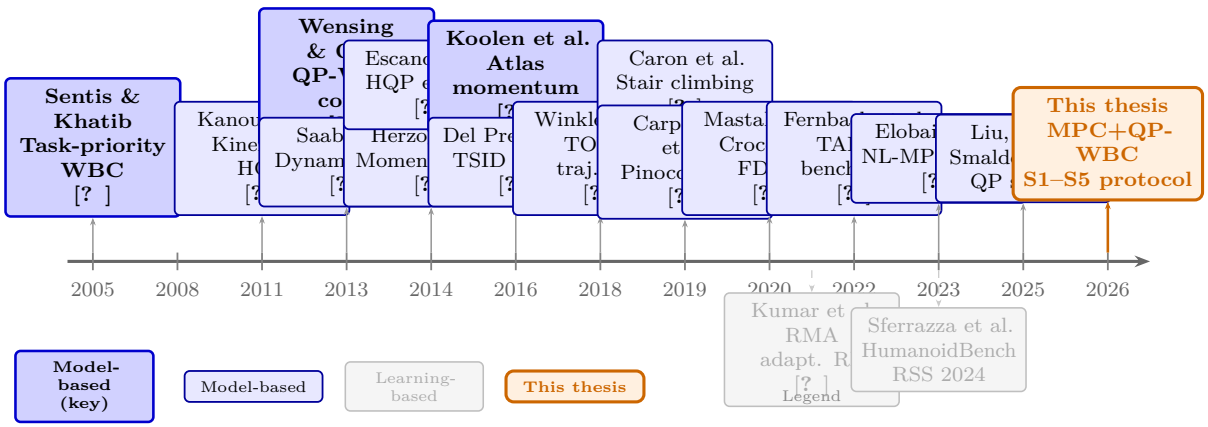

Figure 2.2: Timeline of key contributions to model-based Whole-Body Control for humanoid robots (2005–2026). Foundational works are shown in bold. Learning-based milestones (grey, below the axis) are included for context only. The proposed thesis contribution is indicated at 2026.

2.4 Model Predictive Control for Humanoid Locomotion

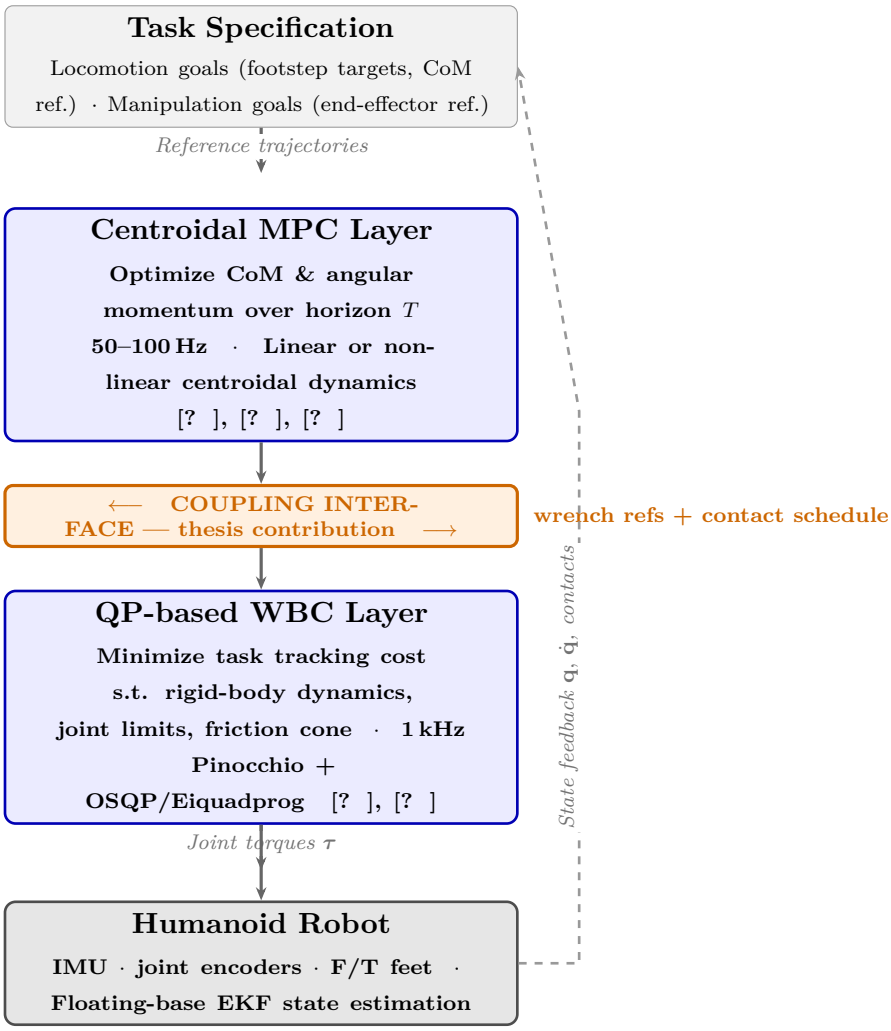

Figure 2.3: Hierarchical planning/execution architecture for model-based humanoid Whole-Body Control. The centroidal MPC layer generates wrench references and a contact schedule at 50–100 Hz; the QP-based WBC layer distributes these to joint torques at 1 kHz. The coupling interface (orange) is the underdocumented boundary identified as a methodological gap in the literature and constitutes a primary contribution of this thesis.

Model Predictive Control (MPC) addresses a complementary problem to Quadratic Program (QP)-based Whole-Body Control (WBC): instead of solving an instantaneous tracking optimisation at each cycle, MPC solves a finite-horizon optimal control problem over a predicted trajectory [?]. For humanoid locomotion, MPC is dominantly applied at the *centroidal* level — optimising $\boldsymbol{p}_{\mathrm{CoM}}$ and angular-momentum trajectories under contact constraints [? ? ?] — because full-body MPC at high frequency is computationally intractable.

Three architectures dominate the post-2022 literature. The first is the *linear / linearised centroidal MPC* lineage descended from the Linear Inverted Pendulum Model, which preserves real-time feasibility (under 10 ms per solve) at the cost of accuracy in highly dynamic regimes [?]. The second is *non-linear centroidal MPC*, exemplified by Romualdi

et al. [?] and Elobaid et al. [?], which retains full centroidal non-linearity at the cost of solve times in the 50–70 ms range on the iCub3 platform [?]. The third is *trajectory optimisation with phase-based parameterisation* [?], which absorbs gait sequencing into the optimisation itself by treating step timings and footholds as decision variables — a structural innovation subsequently adopted in step-adjustment-aware MPC [?].

A separate trajectory-optimisation tradition uses *multi-shooting* and DDP-class algorithms over the full-body dynamics. Crocoddyl [?] is the reference implementation, supporting Feasibility-driven Differential Dynamic Programming (FDDP) over multi-contact problems and computing dynamic manoeuvres (jumps, front-flips) within milliseconds. Crocoddyl is built on Pinocchio's analytical-derivative dynamics [?], which evaluates legged-robot dynamics in approximately 3 μs and is the de facto industry standard.

The interface between MPC (planning layer) and QP-WBC (execution layer) is the *unresolved* element of this literature. The post-2022 sweep confirms it remains open: Liu et al. [?], Wang et al. [?], and Smaldone et al. [?] each propose plausible coupling architectures using mutually incompatible terminology.

Critical Insight

MPC and QP-WBC are not competing methods but layered ones, and the more useful question is not "MPC or WBC?" but "what is the right interface between them?" The current literature offers at least three answers — sequential time-scale separation [?], explicit objective sharing [?], and contact-explicit hierarchical RL+MPC [?] — but no consensus. This is the precise methodological territory the thesis enters in its proposed methodology chapter; Section 2.7 will argue that *naming and unifying* this design space, even before innovating within it, is a defensible Chapter 2 contribution.

2.5 Learning-based Approaches

Reinforcement Learning (RL) approaches to humanoid Whole-Body Control (WBC) have advanced rapidly since 2022 but are treated in this thesis as a *baseline for comparison* rather than a contribution axis. This positioning is required by the thesis scope but also aligns with three substantive observations from the literature.

First, RL-based WBC has demonstrated impressive *agility*: end-to-end learned policies on Cassie and Unitree H1 achieve dynamic walking, recovery from large perturbations, and sim-to-real transfer with domain randomisation [? ? ?]. Second, however, RL approaches generally *forfeit interpretability and constraint guarantees*: they cannot enforce hard friction-cone or torque-saturation constraints by construction, only encourage compliance through reward shaping. Third, the evaluation reporting in the RL literature is markedly more quantitative than in the model-based literature: HumanoidBench [?] reports 27-task success rates on Unitree H1, and PolySim reports a +52.8% sim-to-MuJoCo motion-tracking improvement under multi-simulator domain randomisation [?].

A growing class of *hybrid model-learning* approaches — RL-augmented contact-explicit MPC [?] and FALCON-class force-adaptive loco-manipulation [?] — sits between the two camps. These works are outside the thesis scope but are noted here as evidence that the model-based / learning-based distinction is itself becoming porous.

The thesis adopts a pre-trained RL policy (e.g. for Unitree H1 locomotion) as an *upper-bound reference* baseline, against which the proposed QP-WBC framework is compared on the metrics defined in the thesis's benchmark-metrics specification.

Critical Insight

The decision to scope RL out of the contribution but in to the baselines is methodologically coherent but epistemically uncomfortable: if RL outperforms the proposed WBC on every metric, the thesis must defend its design goals (constraint guarantees, interpretability) explicitly rather than performance-numerically. The response — *RL trades constraints for agility, different design goals* — is sustainable but must be reinforced by demonstrating a *qualitative* advantage of the WBC framework, not merely a quantitative one. Section 2.7 returns to this point.

2.6 Simulation Environments and Toolchains

Simulation choice is non-incidental for a simulation-only thesis: it determines what physical behaviours are reproducible and what claims are defensible. The literature converges on four primary platforms.

Pinocchio [?] is the dominant rigid-body dynamics library: it provides RBDL-class forward and inverse dynamics with analytical derivatives, evaluating in approximately 3 μs per call on legged systems. It is the backbone of Stack-of-Tasks and Crocoddyl [?] and is the de facto standard for Quadratic Program (QP)-Whole-Body Control (WBC) implementations.

Crocoddyl [?] extends Pinocchio for trajectory optimisation under contact sequences, with Feasibility-driven DDP as its solver innovation. Crocoddyl is positioned as the reference toolchain for full-body MPC.

MuJoCo [?] is widely used for contact-rich humanoid Simulation, valued for its stable spring-damper soft-contact model and computational speed. Its principal limitation is that soft contacts diverge from rigid hardware reality in ways that are platform-dependent.

Isaac Sim (NVIDIA) provides GPU-accelerated, photorealistic Simulation with rigid-body dynamics, enabling massively parallel RL training [?]. Its model-based use for WBC remains less documented but is growing [?].

A recent benchmarking effort by Fernbach et al. [?] explicitly compared three control schemes (HQP velocity, TSID acceleration, TSID torque) on TALOS in Gazebo, reporting $\boldsymbol{p}_{\mathrm{CoM}}$ tracking peaks of 5 cm in x and 1.5 cm in y during walking. This benchmark is the closest existing analogue to the evaluation protocol the thesis proposes and is therefore a critical reference point for the methodology and experiments chapters.

Critical Insight

The toolchain landscape is not neutral: choosing Pinocchio + MuJoCo commits the thesis to a specific contact-model debate (soft vs. rigid), while Pinocchio + Isaac Sim commits to a different one (CPU vs. GPU determinism). The selection between MuJoCo and Isaac Sim cannot be deferred without consequence: the literature suggests MuJoCo is better-supported for torque-controlled WBC research [?], while Isaac Sim is better-supported for sim-to-real RL [?]. Because the thesis is not RL-centric, MuJoCo is the more conservative methodological choice and should be confirmed early in the methodology chapter to unblock downstream work.

2.7 Critical Synthesis — Gaps and Positioning

The preceding sections establish three structural facts about the model-based humanoid Whole-Body Control (WBC) literature.

The first is that the dominant architectural pattern is *layered*: a planning layer — centroidal Model Predictive Control (MPC), footstep planner, or trajectory optimisation — supplies references to an execution layer built around a Quadratic Program (QP)-based WBC at torque level, and this split is widely acknowledged as the pragmatic resolution of the centroidal-versus-full-body dynamics controversy [? ? ? ? ?]. Post-2022 contributions — whether focused on heavy-limbed humanoids [?], underactuated platforms [?], computationally efficient footstep planners coupled to WBC [?], or contact-stable force parametrisations under payload [?] — all retain this two-layer architecture. The thesis architecture (centroidal MPC paired with QP-WBC) therefore inherits a consolidated design pattern rather than proposing a new one.

The second fact is that, despite this architectural consolidation, the *interface between the two layers remains underspecified*. Romualdi et al. couple the layers via non-linear centroidal MPC with step timing as an optimisation variable [?]; Elobaid et al. extend this coupling with a contact-stable force parametrisation for payload carrying [?]; Wang et al. embed the MPC cost inside the whole-body planner-controller stack [?]; Smaldone et al. propose compute-budget-aware decoupling at the planner/WBC boundary [?]; Patrizi et al. augment the MPC with learned residual corrections inside a contact-explicit hierarchy [?]. These proposals are plausible in isolation but use incompatible terminology, and no published work surveyed here reconciles them into a common formalism or compares them on a shared benchmark.

The third fact is that the *evaluation protocol for simultaneous loco-manipulation is effectively absent*. The unified MPC formulation of Sleiman et al. for whole-body dynamic locomotion and manipulation [?] is a rare case in which the simultaneous loco-manipulation scenario is exercised under model-based control; recent learning-based loco-manipulation works [? ?] sidestep the constraint-enforcement question entirely. Friction-cone compliance is rarely reported quantitatively because QP-WBC enforces it as a hard constraint by construction [? ?], and task success rates are almost never accompanied by confidence intervals or infeasibility statistics even in the TALOS benchmarking study of Fernbach et al. [?], which is otherwise the closest analogue to a standardised evaluation.

These three facts resolve into three explicit gaps that define the space available to the thesis.

Research Gap

Methodological gap — planner/WBC coupling. No consensus framework couples a planning layer to a Quadratic Program (QP)-Whole-Body Control (WBC) execution layer across different time scales, even as several candidate couplings coexist in the literature without a reconciliation mechanism [? ? ? ?].

Research Gap

Evaluation gap — loco-manipulation protocol. No standardised scenario set, metric set, or reporting convention governs simultaneous humanoid loco-manipulation under Whole-Body Control (WBC). The existing TALOS benchmark restricts itself

to locomotion-only tasks [?], and published whole-body loco-manipulation studies do not share a common evaluation template [?].

Research Gap

Measurement gap — QP infeasibility as silent failure. The model-based community treats QP infeasibility as a silent failure mode rather than a first-class observable, even though every infeasibility event is informative given the hard-constraint formulation [? ?]. No widely adopted reporting protocol tracks infeasibility rates, contact-cycle feasibility, or confidence-intervaled task success.

	Constraint satisfaction (hard)	Computational cost (low)	Multi-contact support	Real-time (1 kHz)	Loco-manip. support	Evaluation protocol
Proposed: MPC + QP-WBC	✓	∼	✓	✓	✓	✓
Kin-based *(baseline)* [?]	×	✓	∼	✓	✓	∼
MPC [?]	✓	×	✓	×	∼	×
Linear centroidal MPC [?]	∼	✓	∼	∼	×	×
Hierarchical QP (HQP) [?]	✓	∼	✓	✓	∼	×
QP-based WBC (single) [?]	✓	∼	∼	✓	∼	×
Priority [?]	×	✓	×	✓	×	×

✓ = satisfied / low / supported ∼ = partial × = not satisfied / high / absent

Figure 2.4: Qualitative comparison of Whole-Body Control approaches across six evaluation dimensions. ✓ = satisfied / low cost / supported; ∼ = partial; × = not satisfied / high cost / absent. The proposed framework (last row, highlighted) scores favourably on constraint satisfaction, multi-contact support, loco-manipulation, and evaluation protocol — dimensions where existing approaches show the clearest limitations. Scores are grounded in the literature surveyed in Sections 2.2–2.6.

Against these three gaps the thesis positions its contribution with deliberate modesty in one direction and deliberate ambition in the other. The thesis does *not* claim radical algorithmic novelty: the planning layer uses established non-linear centroidal MPC formulations [? ?], the execution layer uses a standard weighted QP-WBC [? ?], and the dynamics and optimisation stack reuses mature open-source infrastructure [? ?]. A reviewer expecting a new solver, a new cost formulation, or a new contact model will not find one in the methodology chapter. What the thesis does claim is a *consolidation-and-validation* contribution, which can be stated along three concrete axes. First, one explicit coupling between centroidal MPC and QP-WBC is instantiated in Simulation and documented at a level of detail sufficient for reproduction, closing part of the methodological gap. Second, evaluation is performed on a scenario set S_1–S_5 that includes the simultaneous loco-manipulation case most often omitted in the literature [? ?]. Third, the reporting protocol treats QP infeasibility rates, contact-cycle feasibility, and confidence-intervaled task success as first-class metrics. Taken individually, each of these pieces exists somewhere in the literature; taken together, under rigorous reporting, they

do not. This conjunction is what qualifies the thesis as a master's-level contribution with plausible publication value.

Critical Insight

Chapter 2 itself has two acknowledged limits that should temper the strength of its claims. The first is that its evidentiary base is bounded by the seven verified foundational sources and the twenty-five post-2022 candidates triaged during preparation; papers not retrieved by the targeted search — in particular non-English venues and robotics workshops not indexed by standard paper corpora — may contain counter-examples to the evaluation-absence claim, and the strength of the third gap above is therefore contingent on the completeness of the sweep rather than on an exhaustive proof of non-existence. The second limit is that hybrid model-plus-learning approaches [**?** **?**] are deliberately scoped out, even though their coupling-of-layers strategies are methodologically closest to what the methodology chapter will propose. This exclusion is defensible given the thesis scope, but it leaves open the reviewer question of whether a learning-augmented MPC would subsume the proposed contribution; that question is deferred to the discussion chapter rather than answered here.

Chapter 2 — Composite Critical Insight

Synthesising across Sections 2.1–2.7, the model-based humanoid WBC literature has matured technically — the dynamics, the QP machinery, the centroidal MPC — but stagnated methodologically: evaluation protocols remain ad-hoc, coupling architectures remain unnamed, and loco-manipulation remains underexplored. The thesis is therefore better understood as a *methodological consolidation* contribution than as an algorithmic one, and this framing is more defensible against standard reviewer criticisms than the converse. The remainder of the thesis executes against this framing: the methodology chapter specifies one reference coupling, the simulation-framework chapter instantiates it, the experiments chapter reports against the protocol, and the discussion chapter addresses the limits of the simulation-only validation in light of the sim-to-real asymmetry surfaced in Section 2.5.

Chapter 3

Methodology

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3.1 Framework Overview

Figure 3.1: Overview of the proposed hierarchical WBC framework. The centroidal MPC planning layer generates reference trajectories at 1,kHz, which are tracked by the QP-based WBC execution layer at 1,kHz.

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Critical Insight

3.2 System Model

3.2.1 Floating-Base Dynamics

The humanoid robot is modeled as a floating-base rigid multibody system with n actuated joints. The generalized configuration is $\boldsymbol{q} = [\boldsymbol{q}_b^\top, \boldsymbol{q}_j^\top]^\top \in \mathbb{R}^{6+n}$, where $\boldsymbol{q}_b \in SE(3)$ represents the unactuated base pose and $\boldsymbol{q}_j \in \mathbb{R}^n$ the joint coordinates. The equations of motion are:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{S}^\top \boldsymbol{\tau} + \boldsymbol{J}_c^\top \boldsymbol{f}_c \tag{3.1}$$

where $\boldsymbol{M}(\boldsymbol{q}) \in \mathbb{R}^{(6+n)\times(6+n)}$ is the joint-space inertia matrix, $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) \in \mathbb{R}^{6+n}$ collects Coriolis, centrifugal and gravitational terms, $\boldsymbol{S} = [\boldsymbol{0}_{n\times 6}, \boldsymbol{I}_n]$ is the actuation selection matrix, $\boldsymbol{\tau} \in \mathbb{R}^n$ is the joint torque vector, $\boldsymbol{J}_c$ is the contact Jacobian, and $\boldsymbol{f}_c \in \mathbb{R}^{3k}$ is the contact wrench (k contact points).

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Critical Insight

3.2.2 Centroidal Dynamics

$$\begin{bmatrix} \dot{\boldsymbol{l}} \\ \dot{\boldsymbol{k}} \end{bmatrix} = \sum_i \begin{bmatrix} \boldsymbol{f}_{c,i} \\ \boldsymbol{r}_i \times \boldsymbol{f}_{c,i} \end{bmatrix} + \begin{bmatrix} m\boldsymbol{g} \\ \boldsymbol{0} \end{bmatrix} \tag{3.2}$$

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3.3 QP-Based WBC Formulation

$$\min_{\ddot{\boldsymbol{q}}, \boldsymbol{\tau}, \boldsymbol{f}_c} \sum_i w_i \left\| \boldsymbol{J}_i \ddot{\boldsymbol{q}} + \dot{\boldsymbol{J}}_i \dot{\boldsymbol{q}} - \ddot{\boldsymbol{x}}_i^{\text{ref}} \right\|_{\boldsymbol{W}_i}^2 \tag{3.3}$$

subject to:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{S}^\top \boldsymbol{\tau} + \boldsymbol{J}_c^\top \boldsymbol{f}_c \qquad \text{(floating-base dynamics)} \tag{3.4}$$

$$\boldsymbol{J}_c \ddot{\boldsymbol{q}} + \dot{\boldsymbol{J}}_c \dot{\boldsymbol{q}} = \boldsymbol{0} \qquad \text{(rigid contact)} \tag{3.5}$$

$$\boldsymbol{\tau}_{\min} \leq \boldsymbol{\tau} \leq \boldsymbol{\tau}_{\max} \qquad \text{(torque limits)} \tag{3.6}$$

$$\boldsymbol{f}_c \in \mathcal{FC} \qquad \text{(friction cone)} \tag{3.7}$$

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Critical Insight

3.4 Centroidal MPC Planning Layer

Nulla non mauris vitae wisi posuere convallis. Sed eu nulla nec eros scelerisque pharetra. Nullam varius. Etiam dignissim elementum metus. Vestibulum faucibus, metus sit amet mattis rhoncus, sapien dui laoreet odio, nec ultricies nibh augue a enim. Fusce in ligula. Quisque at magna et nulla commodo consequat. Proin accumsan imperdiet sem. Nunc porta. Donec feugiat mi at justo. Phasellus facilisis ipsum quis ante. In ac elit eget ipsum pharetra faucibus. Maecenas viverra nulla in massa.

Critical Insight

3.5 Contact Modeling and Mode Transitions

3.5.1 Rigid Contact Model

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3.5.2 Soft Contact Model

Etiam pede massa, dapibus vitae, rhoncus in, placerat posuere, odio. Vestibulum luctus commodo lacus. Morbi lacus dui, tempor sed, euismod eget, condimentum at, tortor. Phasellus aliquet odio ac lacus tempor faucibus. Praesent sed sem. Praesent iaculis. Cras rhoncus tellus sed justo ullamcorper sagittis. Donec quis orci. Sed ut tortor quis tellus euismod tincidunt. Suspendisse congue nisl eu elit. Aliquam tortor diam, tempus id, tristique eget, sodales vel, nulla. Praesent tellus mi, condimentum sed, viverra at, consectetuer quis, lectus. In auctor vehicula orci. Sed pede sapien, euismod in, suscipit in, pharetra placerat, metus. Vivamus commodo dui non odio. Donec et felis.

3.5.3 Contact Mode Transition Strategy

Etiam suscipit aliquam arcu. Aliquam sit amet est ac purus bibendum congue. Sed in eros. Morbi non orci. Pellentesque mattis lacinia elit. Fusce molestie velit in ligula. Nullam et orci vitae nibh vulputate auctor. Aliquam eget purus. Nulla auctor wisi sed ipsum. Morbi porttitor tellus ac enim. Fusce ornare. Proin ipsum enim, tincidunt in, ornare venenatis, molestie a, augue. Donec vel pede in lacus sagittis porta. Sed hendrerit ipsum quis nisl. Suspendisse quis massa ac nibh pretium cursus. Sed sodales. Nam eu neque quis pede dignissim ornare. Maecenas eu purus ac urna tincidunt congue.

Critical Insight

3.6 State Estimation

Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus. Morbi purus. Nulla a est sit amet purus venenatis iaculis. Vivamus viverra purus vel magna. Donec in justo sed odio malesuada dapibus. Nunc ultrices aliquam nunc. Vivamus facilisis pellentesque velit. Nulla nunc velit, vulputate dapibus, vulputate id, mattis ac, justo. Nam mattis elit dapibus purus. Quisque enim risus, congue non, elementum ut, mattis quis, sem. Quisque elit.

Critical Insight

3.7 Task Hierarchy and Priority Assignment

Maecenas non massa. Vestibulum pharetra nulla at lorem. Duis quis quam id lacus dapibus interdum. Nulla lorem. Donec ut ante quis dolor bibendum condimentum. Etiam egestas tortor vitae lacus. Praesent cursus. Mauris bibendum pede at elit. Morbi et felis a lectus interdum facilisis. Sed suscipit gravida turpis. Nulla at lectus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Praesent nonummy luctus nibh. Proin turpis nunc, congue eu, egestas ut, fringilla at, tellus. In hac habitasse platea dictumst.

Critical Insight

Chapter 4

Simulation Framework

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4.1 Simulator Selection and Justification

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Critical Insight

4.2 Robot Model

Property	Value
Platform	[TALOS / Unitree H1 / Custom]
Total DOF	[28–32]
Actuation	[Torque-controlled / SEA]
Base sensors	IMU (1,kHz)
Joint sensors	Encoders (1,kHz)
Foot sensors	Force/torque (1,kHz)
URDF source	[URL / repository]

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metus, malesuada sed, condimentum sit amet, auctor a, wisi. Pellentesque sapien elit, bibendum ac, posuere et, congue eu, felis. Vestibulum mattis libero quis metus scelerisque ultrices. Sed purus.

Critical Insight

4.3 Software Architecture

Figure 4.1: Software stack architecture. Real-time loop runs at 1,kHz; MPC runs asynchronously at 1,kHz.

Donec molestie, magna ut luctus ultrices, tellus arcu nonummy velit, sit amet pulvinar elit justo et mauris. In pede. Maecenas euismod elit eu erat. Aliquam augue wisi, facilisis congue, suscipit in, adipiscing et, ante. In justo. Cras lobortis neque ac ipsum. Nunc fermentum massa at ante. Donec orci tortor, egestas sit amet, ultrices eget, venenatis eget, mi. Maecenas vehicula leo semper est. Mauris vel metus. Aliquam erat volutpat. In rhoncus sapien ac tellus. Pellentesque ligula.

Critical Insight

4.4 Control Loop Implementation

```
[H]


    Input: Sensor readings $(\q, \dq, \fc^{\mathrm{meas}})$, MPC reference $\xtask^{\mat
    Output: Joint torques $\torque$
     Estimate floating-base state via EKF: $\hat{\q}_b, \dot{\hat{\q}}_b$
     Detect contact modes from foot force readings
     Update contact Jacobian $\Jc$ and task Jacobians $\J_i$
     Formulate QP \cref{eq:qp-cost}--\cref{eq:qp-friction}
     Solve QP $\rightarrow$ $(\ddq^*, \torque^*, \fc^*)$
     return $\torque^*$
```

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4.5 Validation of the Simulation Setup

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Critical Insight

Chapter 5

Experiments and Results

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5.1 Experimental Setup

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5.2 Evaluation Metrics

Table 5.1: Evaluation metrics (thresholds to be grounded in literature, PQ6).

Metric	**Symbol**	**Definition**	**Target**
CoM tracking error	e_{CoM}	RMSE of $\boldsymbol{p}_{\text{CoM}}$ vs. reference	$< X$ cm
Contact force compliance	η_{FC}	Fraction of time inside friction cone	$> 95\%$
Task success rate	η_{task}	% of trials completing target task	$> Y\%$
QP solve time	t_{QP}	Mean/max per control cycle	$< 1, kHz$
Perturbation recovery	η_{rec}	Recovery rate after impulse	TBD
End-effector error	e_{EE}	Position RMSE during manipulation	$< Z$ cm

5.3 Baselines

1. **Null-space Task-Priority WBC** — classical Sentis & Khatib formulation [**?**]; no inequality constraint handling.
2. **PD Joint Control** — decentralized joint-level PD; no whole-body coordination.

3. **RL Policy** — pre-trained locomotion policy (e.g., Unitree H1 RL); used as agility upper-bound reference only.

5.4 S1: Static Balancing

Sed eleifend, eros sit amet faucibus elementum, urna sapien consectetuer mauris, quis egestas leo justo non risus. Morbi non felis ac libero vulputate fringilla. Mauris libero eros, lacinia non, sodales quis, dapibus porttitor, pede. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Morbi dapibus mauris condimentum nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Etiam sit amet erat. Nulla varius. Etiam tincidunt dui vitae turpis. Donec leo. Morbi vulputate convallis est. Integer aliquet. Pellentesque aliquet sodales urna.

Table 5.2: S1 — Static balancing results.

Method	e_{CoM} (cm)	η_{FC} (%)	t_{QP} (ms)
Proposed WBC	—	—	—
Task-Priority WBC	—	—	—
PD Control	—	—	—

Critical Insight

5.5 S2: Flat-Ground Walking

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Critical Insight

5.6 S3: Stair Climbing

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Critical Insight

5.7 S4: Loco-Manipulation

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Critical Insight

5.8 S5: Perturbation Robustness

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Critical Insight

5.9 Comparative Analysis

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Critical Insight

Chapter 6

Discussion

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6.1 Hypothesis Assessment

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H3 Assessment. Quisque consectetuer. In suscipit mauris a dolor pellentesque consectetuer. Mauris convallis neque non erat. In lacinia. Pellentesque leo eros, sagittis quis, fermentum quis, tincidunt ut, sapien. Maecenas sem. Curabitur eros odio, interdum eu, feugiat eu, porta ac, nisl. Curabitur nunc. Etiam fermentum convallis velit. Pellentesque laoreet lacus. Quisque sed elit. Nam quis tellus. Aliquam tellus arcu, adipiscing non, tincidunt eleifend, adipiscing quis, augue. Vivamus elementum placerat enim. Suspendisse ut tortor. Integer faucibus adipiscing felis. Aenean consectetuer mattis lectus. Morbi malesuada faucibus dolor. Nam lacus. Etiam arcu libero, malesuada vitae, aliquam vitae, blandit tristique, nisl.

Critical Insight

6.2 Contribution Analysis

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Critical Insight

6.3 Limitations

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Critical Insight

6.4 Sim-to-Real Gap Analysis

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Critical Insight

6.5 Open Problems

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Critical Insight

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

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Critical Insight

7.2 Future Work

Near-term extensions.

- Hardware validation on the selected platform (removes simulation-only limitation).
- Quantitative sim-to-real gap analysis via domain randomization.
- Metric threshold grounding from literature-wide meta-analysis (resolves PQ6).

Long-term research challenges.

- Generalized contact transition framework that does not require per-platform hand-tuning (§7.2 Methodological Gap).
- Long-horizon task sequencing under WBC for service robotics scenarios (§7.4 Application Gap).
- Energy-aware WBC formulations for hardware deployment (§7.4 Application Gap).
- Tighter integration of learning-based priors into the QP cost structure while preserving constraint guarantees.

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